A Deep Learning Approach for Drone Detection and Classification using Radar and Camera Sensor Fusion

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***Abstract*—With the growth of Unmanned Aircraft Systems (UAS) technology and the increasing misuse of small UAS (sUAS), the importance of a reliable method for detecting and classifying aircraft from other flying objects has become apparent. The current approaches for detecting and classifying aircraft and other flying objects are primarily based on solutions that rely on a single sensor, either visual data features or micro-Doppler extraction from radar data. However, these methods may have limitations when it comes to detecting objects at greater distances or in challenging weather conditions. To address the problem, the paper proposes a joint classification network based on radar and camera fusion. The radar network extracts the Spatio temporal features from the radar track and the camera network extracts the deep, complex features from the image. A synchronized radar and camera data is established using multiple field trials during different times of the year. The radar classification using a combination of IMM filters and RNN, the camera detection and classification using YOLOv5, and the combined joint classification network are evaluated on the field dataset. The experimental results greatly increase the classification performance for drones and birds, respectively, to 98% and 94%. This is especially true in situations when a single sensor would struggle to offer reliable classification. The system can accurately classify drones while reducing false alarms caused by other objects, such as birds.**

***Index Terms*—UAS, Detection, Classification, Radar, PTZ, Fusion**

I. Introduction

Unmanned Aerial Vehicles (UAVs) have gained immense popularity in recent times due to their numerous applications such as freight transportation, emergency response, firefighting, inspection, and photography [1]. However, their unauthorized use has led to concerns such as privacy infringement, disruption of air traffic, damage to infrastructure, and harm to individuals [2], [3]. To combat these issues, extensive research has been conducted in the field of UAV detection and identification [4]–[6]. Various detection methods have been proposed in the literature, including Radio Frequency (RF) sensor-based wireless communication detection, sound detection of UAV propellers and motors, and visual or infrared camera-based detection methods relying on daylight reflection or infrared radiation. In addition, radar-based detection has also been demonstrated in ground-based and airborne configurations [7], [8]. However, individual sensors may have varying capabilities, and may not always be sufficient to ensure reliable target classification [9].

Recent advances in deep learning have revolutionized the field of computer vision, enabling the development of highly accurate and efficient object detection and classification algorithms [10]. Convolutional neural networks (CNNs), in particular, have achieved remarkable performance on various visual recognition tasks, including drone detection and classification. However, CNNs rely on large amounts of labeled data, which can be difficult to obtain for drone-related tasks, especially in complex environments. Moreover, the applications such as CNN based object detection or classification are susceptible to occlusion, lighting changes, and other visual disturbances, which can affect their robustness and reliability.

To address these limitations, researchers have proposed to combine camera-based CNNs with radar-based algorithms in a deep fusion framework, which can leverage the strengths of both modalities and enhance their complementary capabilities. With the promising results of camera-radar fusion in autonomous car applications [11], there has been increasing interest in applying this approach to other domains, such as drone detection and classification. In this paper, we explore the potential of camera-radar fusion for improving the accuracy and robustness of drone detection and classification systems. By combining the strengths of both modalities, we aim to enhance the complementary information provided by radar and camera sensors, and to develop a more effective and reliable solution for drone security.

In this paper, we present a summary of related work in UAV detection and identification, followed by a description

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of the experimental setup and the dataset used. We then describe the combined fusion algorithm employed, followed by the experimental results. Finally, we conclude the paper in Section V.

II. Literature Review

Typically, drone detection systems are categorized into four groups based on the sensing techniques employed, namely sound detection, RF detection, radar detection, and optical detection [12]. The use of deep learning methods has made visual detection a prevalent method for detecting drones [13]–[15]. Radar is the most commonly employed technique for detecting airborne objects. But it is difficult to detect drones due to low speed, low flying height and similar radar signatures as birds. Nevertheless, several works have explored the use of micro-Doppler characteristics of drones [16]–[19]. Another method that is frequently used is RF-based detection [20]–[22]. Despite many advantages, RF-based detection can not work for autonomous drones. Authors in [23]–[26], have also investigated acoustic-based detection methods. But, none of these methods can solve the drone detection problem in all situations. Several strategies utilizing sensor fusion have recently been developed to overcome the limitation of individual approaches and achieve high accuracy and reliability in detecting tasks.

In the literature, there are primarily two types of techniques for sensor fusion-based UAV detection. In the first approach, a pair or more sensors identify drones simultaneously. Later, the output of the individual sensors is combined either by fusing their extracted features or fusing their detections to generate the final decision. [27] is one of the recent works where authors have used multiple optical and acoustic sensors for UAV detection. The features are extracted from both sensors and sent to a classifier for detection. Another piece of work in [28] focuses on multiple sensors (visible, infrared, acoustic) fusion for UAV detection and classification. Similarly, the authors [29], [30] presented a radar and camera fusion for UAV detection. The radar detection triggers the camera to confirm the final classification. In [31], the author explored the RF and optical sensor fusion by extracting RF signal features and image features by passing them to feed forward neural network and convolutional neural networks, respectively. The final detection is generated using another deep neural network that takes the extracted features as input.

All of the preceding work on sensor fusion-based drone detection and classification has limited capability in terms of working distance that is less than 200 m ([27]–[30]), weak performance ([31]), the small size of the dataset and used micro-Doppler based features for radar. In this work, long distance (up to 1.2 km) UAV detection and classification system is presented and tested in the field.

Our contributions are as follows

- In this paper, we have presented a real-time radar-camera fusion system for small UAVs detection and classification.
- We collected comprehensive radar and camera data of flying drones, birds and other ground objects at various distances up to 1.2km.
- With several field trials, we have shown good classification performance above 90%.

III. Methodology

A. Experimental setup

The radar-camera system setup for field data collecting and testing is shown in Figure 1. The experiment was conducted in a few places close to Laval, Quebec. The system consists of FMCW radar and high definition pan tilt zoom (PTZ) camera. The radar and PTZ are coordinated, and the PTZ is automatically cued to the target that the radar has identified. Typically, the target track id, x, y, z, range, velocity, and RCS are included in the radar output. With the assistance of the expert monitoring the radar and camera screens, all of the data gathered during field experiments were carefully annotated.

Our experiments were conducted using stationary sensors, but deploying them on a moving platform like an aircraft would present challenges such as variations in captured data due to motion, perspective changes, and varying lighting conditions. Adapting our approach for mobile platforms would require addressing these challenges through techniques like image stabilization, motion compensation, and robust feature extraction methods. Additionally, higher processing requirements and real-time data transmission limitations from a moving platform would need careful consideration for successful implementation. While not the primary focus of this paper, these aspects hold potential for future developments.



Fig. 1. A radar and camera setup on the field for UAV detection and classification.

B. Dataset

1) Radar data: The radar dataset comprises of 9138 UAV, 1369 Birds and 1224 other ground targets (car, human) radar

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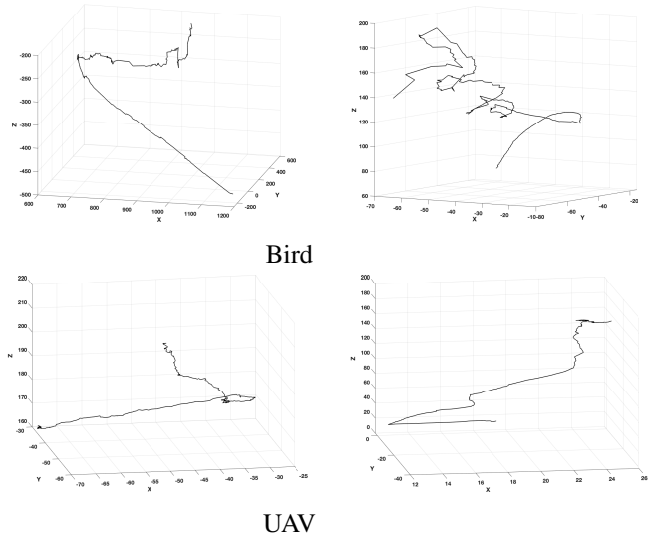


Fig. 2. A sample of real trajectories of birds and UAV recorded by radar.

tracks. Each track is of 5 sec duration. The samples are generated at a sampling frequency of 10 samples/sec by a radar. The dataset contains target track ID, Azimuth, Elevation, Range, X, Y, Z, V_x, V_y, V_z and RCS information. All the tracks are manually annotated by experts with the help of the PTZ camera. Some sample radar tracks of UAV and bird targets are shown in Figure 2.



Fig. 3. Some sample snapshots of the optical data collected from the field trials

2) Optical data: For optical, we collected around 85 videos with 207,663 frames, out of which 154,089 of them have a UAV object, 5,200 of them have OFO (other flying object), and 48,745 of them are background frames without any objects. All of the frames were manually annotated for training the AI models. Table I provides min, mean, and max of the bounding box (BB) size for a better insight into the dataset.

TABLE I
STATISTICS OF BBS FOR OPTICAL DATA (VIDEO RESOLUTION 1920×1080)

Class		Min ($W \times H$)	Mean ($W \times H$)	Max ($W \times H$)
UAV		5×5	50.71×34.06	611×259
	Normalized	0.0029×0.0047	0.0266×0.0321	0.3182×0.2400
OFO		6×2	33.75×24.30	75×81
	Normalized	0.0032×0.0027	0.0179×0.0230	0.0394×0.0756

Figure 3 shows some sample snapshots of the PTZ collected over multiple field trails.

The DJI Phantom 2, Phantom 3, Phantom 4, Phantom 4 Pro, DJI Inspire, and DJI Mavic are the models of UAVs that were used during the trails.

C. Radar and Camera sensor fusion for classification

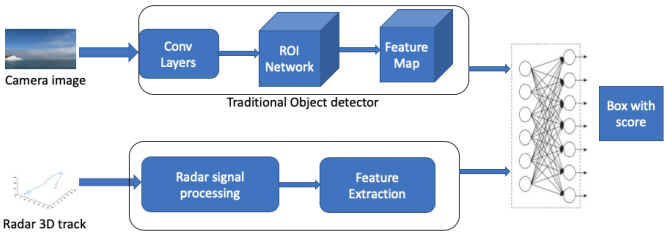


Fig. 4. A general sensor fusion architecture using radar and camera features fed into a classifier system

The goal of this section is to present a technique for target detection and classification by combining data from both radar and camera sensors. Figure 4 illustrates the general architecture of the sensor fusion method that employs camera and radar features for classification. In our system, radar serves as the primary source of object detection and provides an initial classification. After the initial classification from radar, both the optical sensor and radar work together to confirm the final classification.

In Figure 5, we developed a target classifier that extracts spatio-temporal and image features by adapting the structure of the CNN and LSTM classifier. The classifier consists of two branches, namely radar tracks processing and image processing, to handle the differences between time-series point targets and image data. The two branches get inputs with dimensions of (w, h, 1) for track length and total number of features, and input image of (256, 256, 1), respectively. For radar, we used our algorithm that integrates target kinematics and tracking filter features for initial classification [32]. Further, we used our single-stage UAV detector and classifier, YOLO v.5 for image features extraction [33]. The fusion block receives input from the detected bounding box, its feature map, and the feature vector from the radar branch. After obtaining the features from both sensors, the FC (fully connected) layer is

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employed in the fusion block to generate fused features. The classification scores are then computed using the SoftMax Loss function and the fused features. This allowed us to integrate camera data with radar data for more accurate target detection and classification.

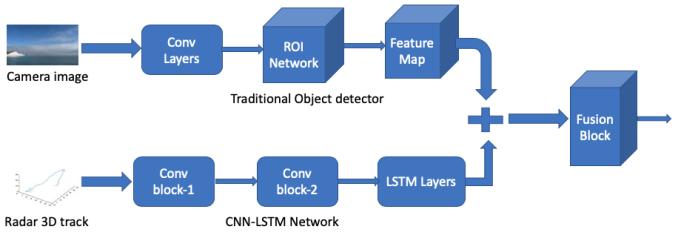


Fig. 5. A deep learning based Radar and Camera sensor fusion classifier.

IV. Results and Discussion

A. Radar only classification

The radar classifier is trained with approximately 12.6 hours of UAV data, 1.9 hours of bird data and 1.7 hours of other ground targets (humans, animals, cars, trucks) data. Figure 6 show the confusion matrix of the decision tree classifier. We can still observe the effect of unbalanced data on the classifier performance. The false positive rate for the birds is 55% and very poor detection for ground based objects. One way to improve the birds' detection is to collect more radar and camera data by doing more experiments during warmer periods of the year. For ground based targets, we can solve the problem with radar cross section for large objects like cars, trucks, but there is some overlap in radar cross section values for humans, animals and low-flying drones. This requires more radar data of humans and other ground based objects.

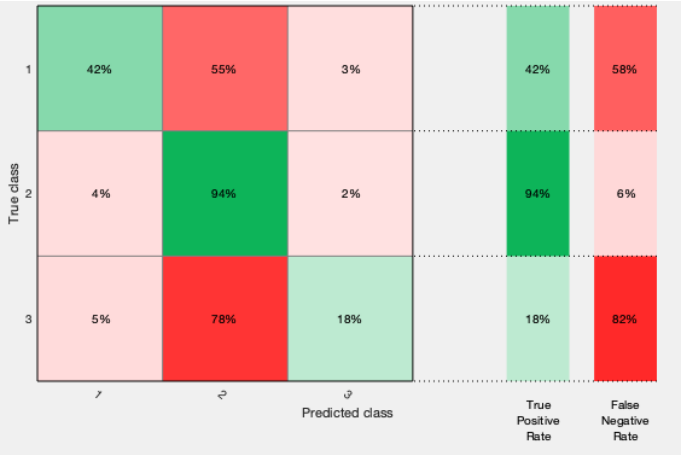


Fig. 6. Confusion matrix for radar only classification, Class-1 Bird, Class-2 UAV, Class-3 Others.

B. Optical only classification

Compared to the confusion matrix, the performance metric for the object detector and classifier is slightly different. The accuracy of object detectors is frequently assessed using the statistic known as AP (Average precision). Its foundations include recall, precision, the confusion matrix, and the idea of intersection over union (IoU).

TABLE II
Performance of different Object detection methods for UAV detection and classification

	Algorithm	AP	AP 0.5
Our custom dataset	FastRCNN	0.557	0.84
	YOLOv3	0.51	0.89
	DETR		
	YOLOv5	0.59	0.91
	YOLOv7	0.58	0.87

Table II summarises the results of several objection detection algorithms trained using our custom dataset. Modern object detection methods like YOLOv5 and YOLOv7 increase AP while also accelerating inference, which is important for real-time applications. Figure 7 displays the model's inference findings based on fresh test data gathered in the field. Even though the model has not seen any images like this during training, it is clear that the model is capable of detecting UAVs with payloads. For scenarios with background trees, the model has trouble.

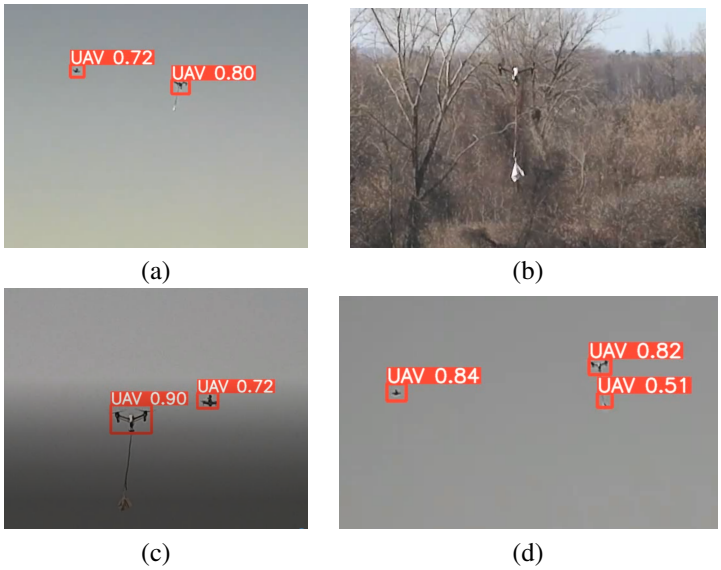


Fig. 7. Performance of visual detection on new data collected for the project.

C. Radar and optical combined classification

Figure 8 shows the radar, optical and combined classification scores for one such track from the experiment logs. We can observe that optical sensor helps us during initial data collection when radar gives a low score. Similarly, radar helps as well when the optical system drops the frame and loses the target. Overall, Table III shows the confusion matrix after

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radar and camera fusion. In this work, the ”other” class refers to ground moving objects such as cars, people, or animals. By focusing on drones and birds in Table III, we aim to emphasize the challenge of distinguishing between these two specific categories that share similarities in flying behavior, radar signatures, and sometimes even appearance. Moreover, it is worth noting that optical sensors based deep learning algorithms are well-established in the literature for classifying ”other” objects, including ground moving ones. By omitting the ”other” class, we can better highlight the performance of our system in its primary objective.

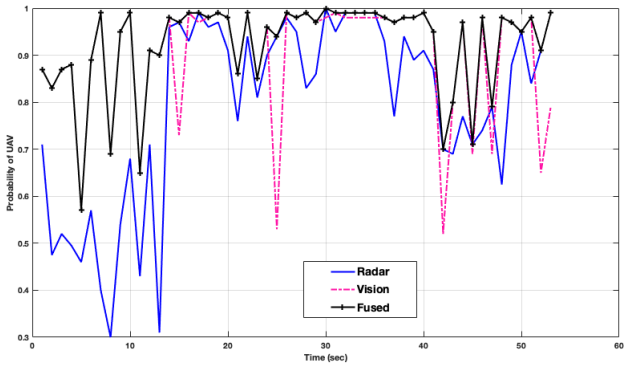


Fig. 8. Comparison of radar, optical and combined classification of an UAV track.

TABLE III
CONFUSION MATRIX FOR COMBINED RADAR AND CAMERA DRONE CLASSIFICATION

	Predicted drone class	Predicted bird class
True drone class	98.4%	1.6%
True bird class	6%	94%

V. CONCLUSION AND FUTURE WORK

In conclusion, the paper presents a novel approach for drone detection and classification using a fusion of radar and optical sensors. The proposed system overcomes the limitations of single-sensor systems and enhances the overall classification performance. By combining the strengths of each sensor, the system is able to accurately detect and classify drones even in complex and dynamic environments.

The results obtained from the real data demonstrate the effectiveness of the proposed approach. The fusion of radar and optical features significantly improves the classification performance, particularly in scenarios where a single sensor would struggle to provide accurate results. The system can accurately classify drones while reducing false alarms caused by other objects, such as birds.

In the future, the proposed approach can be further improved by incorporating more sensors such as acoustic sensors, which can help detect drones that are not visible to radar or optical sensors. Additionally, the system can be trained with more diverse data sets to further enhance its performance in various environments and under different weather conditions. The proposed approach has numerous practical applications in various industries, including security and surveillance, agriculture, and wildlife management. It can help in detecting and preventing potential threats and hazards posed by drones, ensuring the safety and security of individuals and property.

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